

Network Analysis of the Enron Email Dataset

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A network analysis of the famous Enron Email dataset, consisting of network pruning to create a new subset, calculation of relevant values, and corresponding plots for the analysis, including community detection.

CCS Concepts: • **Computing methodologies** → **Network science**.

Additional Key Words and Phrases: Network, Centrality, Community Detection, Clustering, Small-world, Hubs

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1 Introduction

In this report, we will be conducting an analysis related to the famous Enron Email Dataset <https://snap.stanford.edu/data/email-Enron.html>.

Our objective is to perform a thorough network analysis consisting of the creation of a subset, calculation, and plotting of relevant data, followed by the conclusions we will take from them, and community detection and a representation of the community network, with nodes representing the communities of Enron employees rather than just a single employee.

To conclude this report there will be a detailed interpretation of all the data acquired and presented throughout this report relating to network properties and conclusions we can take from them.

2 Relevant Literature Overview

We ran into a problem early into our research, there is a lack of relevant literature regarding simple network analysis of the Enron Email Dataset, while there are plenty of papers on the topic, they tend to branch into more complex analysis, for example regarding the change of network structure and communication between nodes before and after the Enron scandal came to light and focus much more on the identity of the nodes and their positions within the company, rather than the more basic network analysis we will be doing.

This problem isn't too big since if there is no relevant literature to compare our guidance is to proceed as planned and simply not compare data since we have nothing to compare, however after seeking feedback from the professor we decided to still attempt to compare the data to relevant data if we are able to find it.

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3 Subset Creation

To start we needed to prune the full dataset, due to its size and complexity it was hard to visualize properly.

The original dataset consisted of 36692 nodes and 183831 edges.

3.1 K-Core Decomposition

For this task we decided to go with K-Core Decomposition, with the goal of removing the more sparse or less connected nodes and isolating the most important cores of the dataset, which are more heavily connected and form robust communities.

To choose our K we decided to set that we wanted to retain approximately 30% of our subset so we would still have at least 10k nodes to work with, so $K = 5$ resulted in a subset closest to what we were looking for.

3.2 Resulting Subset

The resulting subset after the K-Core Decomposition consisted of 11648 nodes and 138578 edges, however for the purposes of our analysis we wanted our subset to be connected, so we isolated the largest core component and removed the rest.

The result is a connected subset with 11538 nodes and 138245 edges.

4 Subset Analysis

Now with the subset ready we are ready to begin our analysis, it will consist of calculating some initial metrics, such as the density, the average degree, etc., proceeded by the calculation and plotting of the centralities, the clustering and finally by doing the community detection.

All this was done in a Jupyter Notebook which can be found fully executed inside our public GitHub repository <https://github.com/Emily-Case/ADC-Group-8>.

4.1 Initial Metrics

The following are the initial metrics of the network:

Table 1. Initial Metrics

Name	Calculated Values
Nodes	11538
Edges	138245
Density	0.0021
Average Degree	23.96
Average Shortest Path	3.41
Diameter	9

As we can see our network has a very low Density at a value of 0.0021, such a low value is expected, considering this is a real world network, it indicates that our network is very sparse and structurally complex, with global connectivity being upheld by hubs and bridges.

The same can be seen regarding the average degree, at approximately 24 it is quite small compared to the size of the network, it helps reinforce the previous conclusion that this is a very sparse network, but it is also normal and expected considering the size and the fact it is a real world network.

The following two metrics are very relevant to each other, as we can see with an average shortest path of 3.41 any node can reach any other node with 3-4 steps, this is very small, especially given the large size of our network, it means the network is well connected and easily navigable, added to this we also know its diameter is 9, meaning the longest shortest path between any two nodes of the network is 9 steps, this is expected given the very small average shortest path, these two metrics show the network is likely to be small-world.

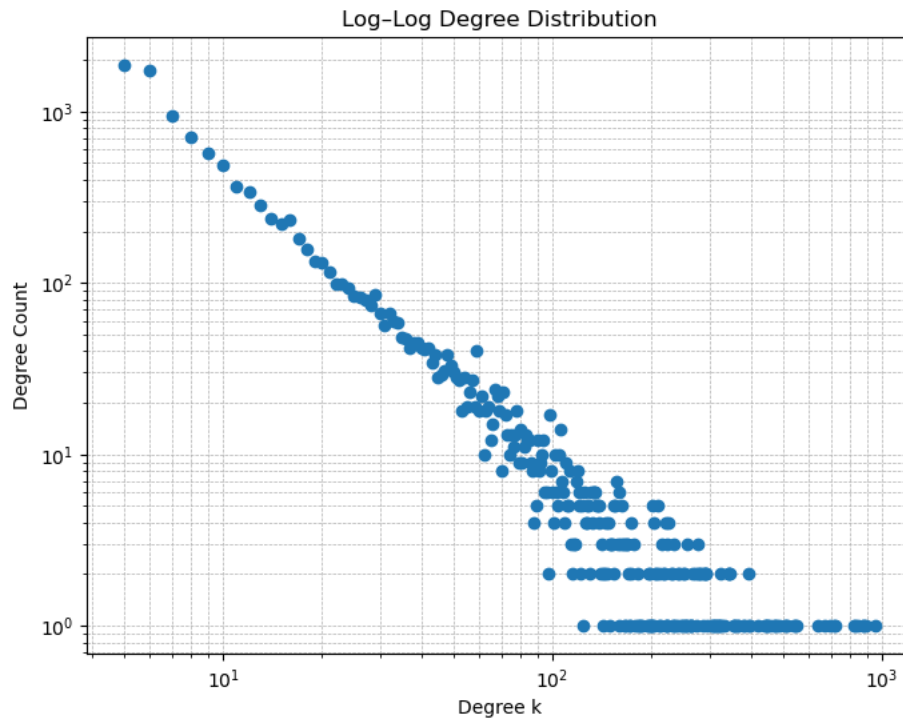


Fig. 1. log-log degree distribution of the network.

Here we can see the log scale degree distribution of the network, which aside from showing the expected behaviour, an almost straight line in log scale that towards the end presents horizontal stripes with a very high number of nodes with small degrees, it also shows the hubs of the network, as we can see by the very small number of nodes with high degrees.

As we can see the current metrics and analysis support the small-worldness of the network.

4.2 Centralities

Following are the plots for the centralities of the network (degree, betweenness, closeness, eigenvector) and the corresponding analysis.

4.2.1 Degree Centrality. Plot:

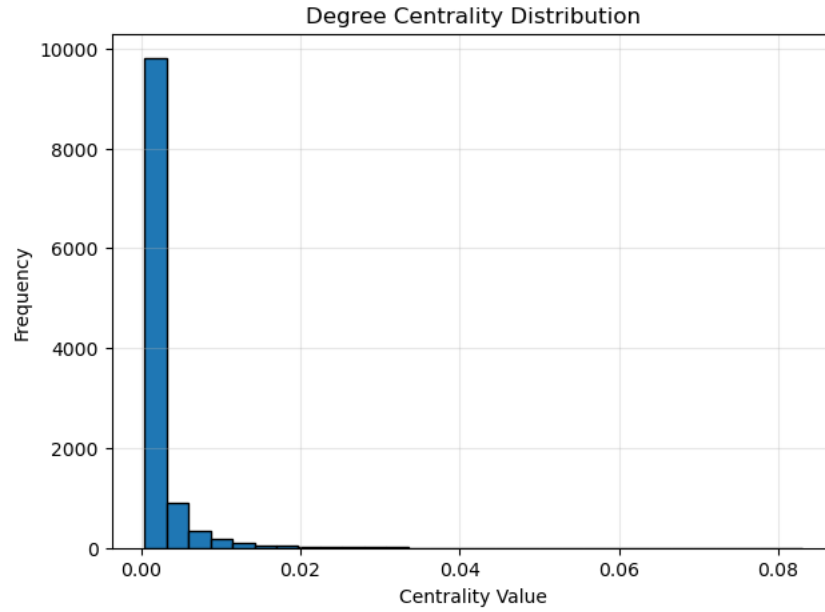


Fig. 2. degree centrality distribution of the network.

As we can see this distribution is extremely skewed, with the overwhelming majority of nodes having very low centrality (below 0.005). This indicates that most nodes have a low degree, while a small number of hubs exhibit significantly higher connectivity, with a maximum degree centrality around 0.08–0.09. This reflects a heavy-tailed network structure, consistent with small-world behavior, where a few highly connected nodes dominate the network topology.

4.2.2 Betweenness Centrality. Plot:

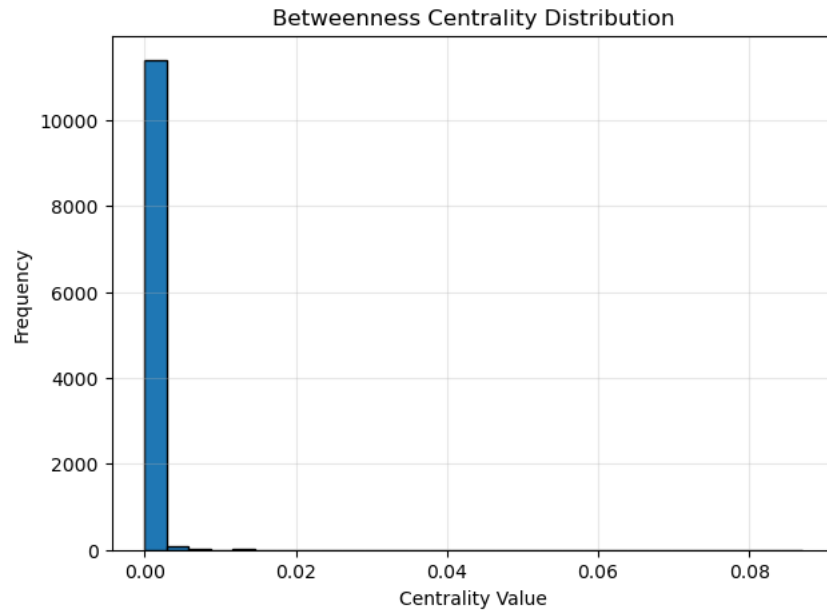


Fig. 3. betweenness centrality distribution of the network.

As we can see this distribution is extremely skewed, with the large majority of nodes having values near zero, and a very small number acting as structural bridges. The highest betweenness centrality is 0.08, meaning that the top node is involved in approximately 8% of all shortest path traffic, or roughly 4–5 million shortest paths. This indicates a strong but not overwhelming broker role, consistent with a small-world network containing multiple hubs and inter-community connectors.

4.2.3 Closeness Centrality. Plot:

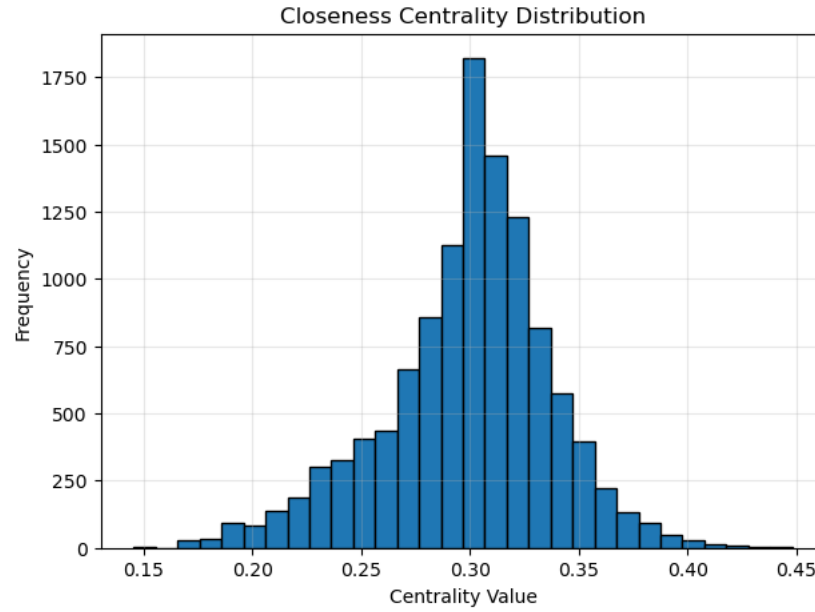


Fig. 4. closeness centrality distribution of the network.

As we can see this distribution is approximately a normal distribution, centered around 0.32. This indicates that most nodes have similar average distances to the rest of the network, consistent with the small-world property (average shortest path length of 3.41). Only a very small number of nodes achieve significantly higher closeness values around 0.45. These nodes represent the hubs that are, on average, one step closer to all other nodes compared with typical nodes. This is typical for large small-world networks where path lengths are highly compressed.

4.2.4 Eigenvector Centrality. Plot:

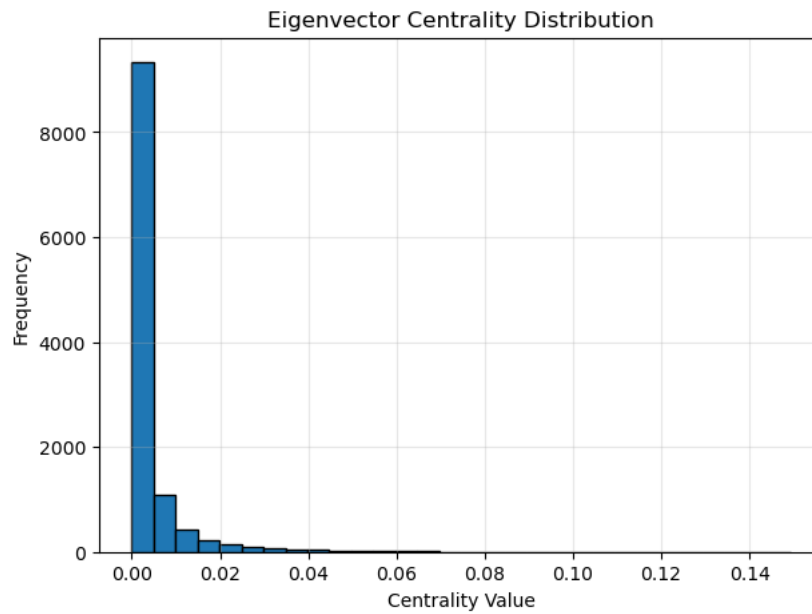


Fig. 5. eigenvector centrality distribution of the network.

As we can see this distribution forms a single, strongly skewed peak, indicating that while the majority of nodes have low influence, a small set of highly connected and well-positioned hubs attain significantly higher centrality values (up to 0.14). This pattern aligns with small-world behaviour, as the most central nodes derive their importance from being connected to other highly connected nodes, forming a tightly interlinked core that helps keep path lengths short throughout the network.

4.3 Clustering

Considering the analysis we have done previously and the small-world behaviour the network has presented, we expected the clustering of this network to be high, which proved to be correct, with an average clustering coefficient of 0.6, a very high value considering the network, it indicates the presence of very dense communities and all but confirms the small-worldness of the network.

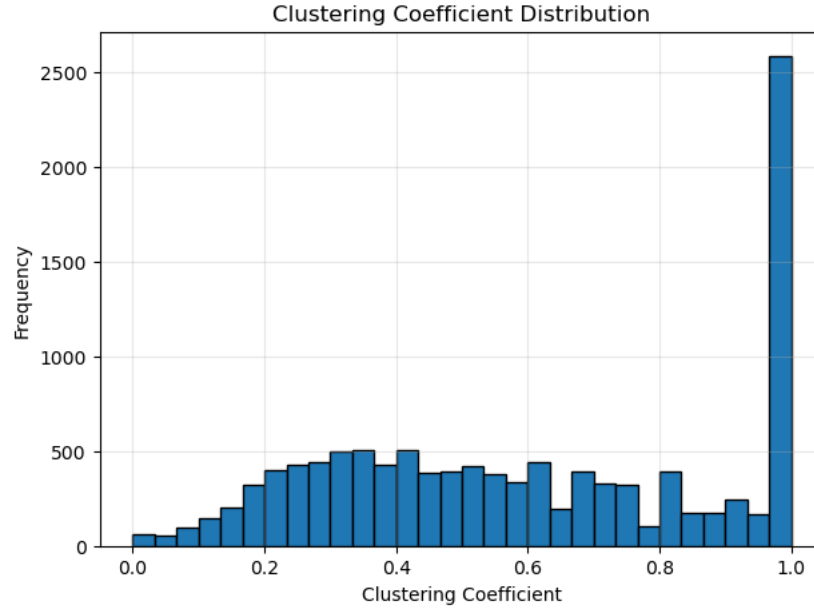


Fig. 6. clustering coefficient distribution of the network.

The clustering plot makes it even more evident, with most nodes having a coefficient between 0.2 and 0.8, but then we have a peak at 0.95 to 1.0 of about 2500 nodes that are as connected as possible and form very dense communities.

4.4 Community Detection

For community detection we decided to use Louvain Community Detection, the reason for our choice was the large size of our network and for its speed, however since Louvain has a degree of randomness to optimize our community detection we run it multiple times and pick the best modularity.

In our detection we reached a best modularity of 0.55, corresponding to 80 communities, we then wanted to represent the full network colouring each node to its community, however the size of the network led to performance and visual issues, so we opted for a different approach, community based visualization, where each node now represents a community with their size proportional to the number of nodes in the community and the edges weighted by inter-community connections.

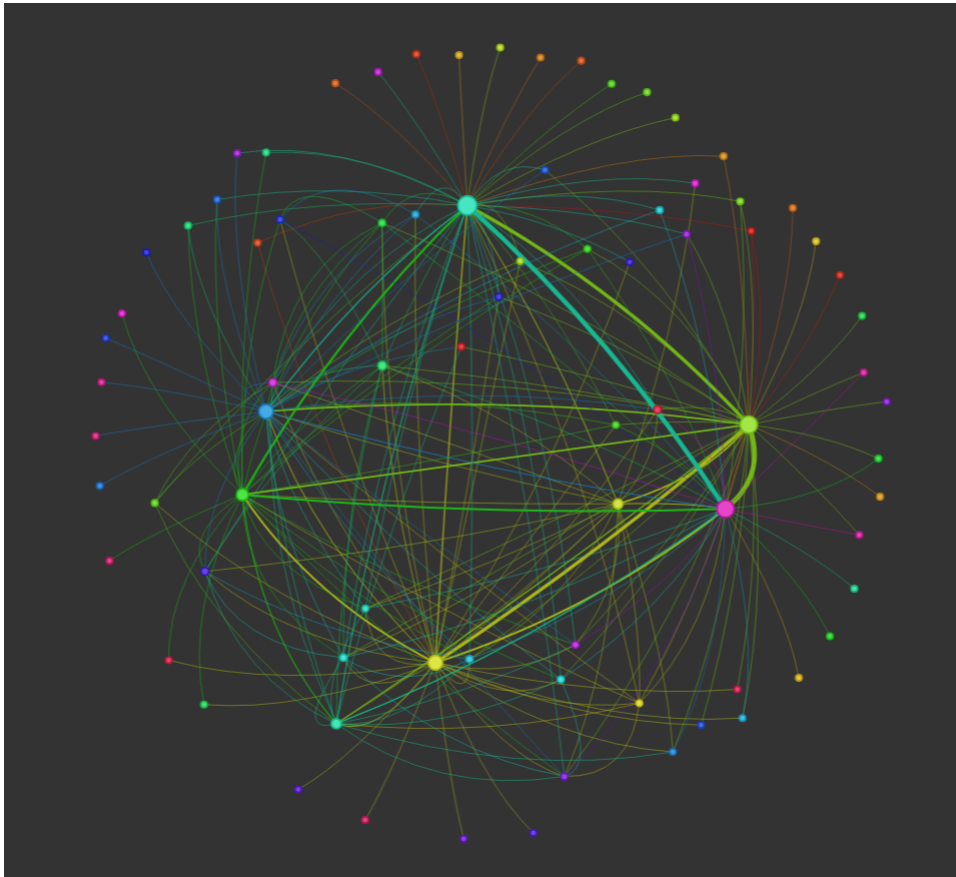


Fig. 7. network representation of the communities found in the network.

As is visible in the image, the network corresponds to what we expected, a small group of 6 bigger communities that are strongly connected between each other and work as hubs that connect smaller communities around the outskirts to the rest of the network.

5 Comparison Tables

Table 2. Metrics Comparison

Name	Our Calculated Values	Relevant Paper 1[2]	Relevant Paper 2[1]
Nodes	11538	36692	33696
Edges	138245	183831	180811
Density	0.0021	N/A	N/A
Average Degree	23.96	10.02	10.73
Average Shortest Path	3.41	N/A	3.99
Average Clustering Coefficient	0.6	0.41	0.71
Diameter	9	12	13

As we can see the result of our pruning to create our subset led to the average degree doubling from the original dataset, we see a marginal increase in the average shortest path and a decrease in the diameter, all of it indicating the network became more connected and easier to traverse, however when it comes to the average clustering coefficient we have differing results, showing a market increase from the first paper and a slight decrease from the second paper.

6 Conclusions

We were able to find behaviours and metrics in our subset that indicate to it being a small-world, such as it's diameter, average shortest path, clustering coefficient, density and the degree distribution.

The subset behaved as expected from our knowledge of the original dataset, with the major differences being direct consequences of our pruning methods, which led to a subset consisting of the node with more connections, since we removed a very large amount of the weakly connected nodes of the network.

To conclude, we had a good experience analyzing this subset as it is a great example of a real life network with a defined hierarchy and organic communities formed around influential and well connected nodes.

References

- [1] Anirban Dasgupta Michael W. Mahoney Jure Leskovec, Kevin J. Lang. 2008. *Community Structure in Large Networks: Natural Cluster Sizes and the Absence of Large Well-Defined Clusters*. Retrieved December 06, 2025 from <https://doi.org/10.48550/arXiv.0810.1355>
- [2] Zhe Chen XiaoJuan Wang, XinXin Zhang. 2011. *The Topological Analysis for Undirected Complex Network Based on Enron Email Database*. Retrieved December 06, 2025 from <https://ieeexplore.ieee.org/document/6154043>